Loan Default Prediction

Comparing Classic ML and Neural Networks under Class Imbalance

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Home Credit Default Risk · Kaggle

Problem & Motivation

The task

Predict the probability that a loan applicant defaults (binary classification, $\texttt{TARGET} \in \{0, 1\}$).

Why class imbalance matters

- Only $\sim$**8.07%** of applicants default — the data is severely imbalanced.
- A "predict-no-default" baseline scores 92% accuracy with **zero recall**.
- Even a model with ROC AUC $= 0.745$ can classify *zero* defaulters at the 0.5 threshold.

Research question

Do classic shallow learners and neural networks respond differently to class imbalance handling?

Dataset · Home Credit Default Risk

Size

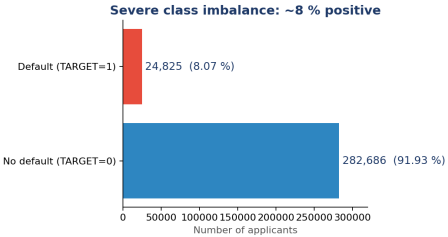
- Train: **307,511** applicants $\times$ 122 features
- Test: 48,744 applicants (Kaggle hidden labels)

Class imbalance

- Default rate $\approx$ **8.07%** (24,825 / 307,511)

Strongest signals

- `EXT_SOURCE_1/2/3` external credit scores ($|\rho| \approx 0.16$–0.18)
- `DAYS_EMPLOYED` = 365,243 sentinel flags 18% (unemployed)
- Building-info columns are 50–70% missing



Methodology · Shared Pipeline

Preprocessing (identical for all five models)

- **80/20 stratified** train–val split (`random_state = 42`)
- Drop `SK_ID_CURR`; keep IDs only for the submission file
- Mean imputation for numeric NaN; one-hot encode 16 categoricals → **244 features**
- `StandardScaler` fitted on train only (essential for LR / MLP)

Tuning & validation

- `GridSearchCV` with `PredefinedSplit` (single train→val fold, no random k-fold)
- Scoring: `roc_auc`; best parameter setting picked per model
- CV CSVs saved to `result/home_credit/cv_results/GridSearchCV/`

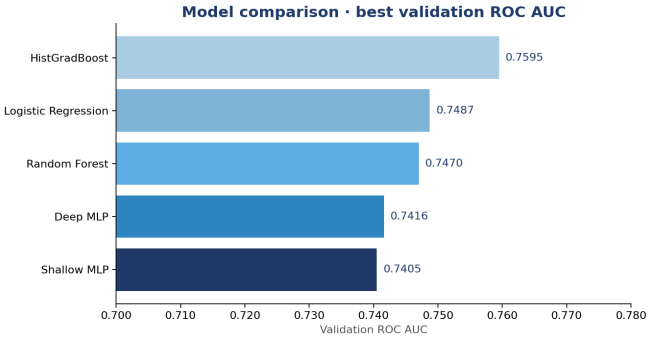
Five Models Compared

Model	Family	Owner	Tuned hyperparameters
Logistic Regression	Linear	Nathaniel	`C`, `tol` | `class_weight=balanced`
Random Forest	Tree (bagging)	Nathaniel	`n_estimators`, `min_samples_*` | `class_weight=balanced`
HistGradientBoosting	Tree (boosting)	Nathaniel	`learning_rate`, `max_iter`, `min_samples_leaf`
Shallow MLP – hidden $(64,)$	Neural network	Minwoo	`alpha`, `learning_rate_init`
Deep MLP – hidden $(128, 64, 32)$	Neural network	Minwoo	`alpha`, `learning_rate_init`

Why this lineup?

- Spans the curriculum: linear $\rightarrow$ tree (bagging + boosting) $\rightarrow$ shallow + deep neural networks.
- Same preprocessing, same train/val split, same scoring $\rightarrow$ *fair* comparison.

Combined Validation ROC AUC



Model	Val AUC
HistGradBoost	**0.7595**
Logistic Reg.	0.7487
Random Forest	0.7470
Deep MLP	0.7416
Shallow MLP	0.7405

Tree boosting wins by $\sim$1 AUC point — consistent with the literature on tabular credit data.

Neural Network Track

Architecture

- **Shallow MLP** — hidden $(64,)$ $\sim$15.7K parameters
- **Deep MLP** — hidden $(128, 64, 32)$ $\sim$42K parameters
- ReLU activation, Adam optimizer, `early_stopping = True`

Hyperparameter tuning

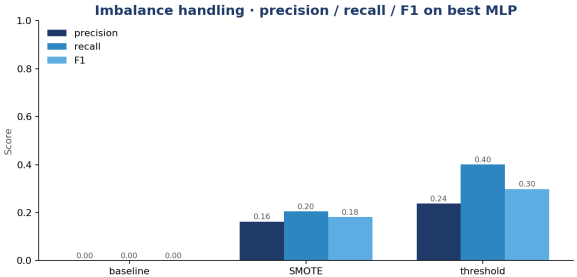
- `alpha` (L2 reg.) $\in \{10^{-4}, 10^{-3}\}$, `lr_init` $\in \{0.001, 0.005\}$
- **Best** for both: $\alpha = 10^{-3}$, `lr` $= 0.005$

Findings

- Shallow 0.7405 vs Deep 0.7416 $\Rightarrow$ depth adds $\sim$ 0.001 AUC (noise).
- NN ranks 4th/5th overall — bottleneck is the *feature set*, not the model.

Imbalance Handling on the Best MLP

Method	Threshold	ROC AUC	PR AUC	Precision	Recall	F1
Baseline	0.500	0.745	0.222	0.00	0.00	0.00
SMOTE	0.500	**0.648**	0.134	0.16	0.21	0.18
Threshold tuning ⋆	0.164	0.745	0.222	0.24	**0.40**	**0.30**



Headline

- Baseline: AUC 0.745, recall **0%** (no prediction crosses 0.5).
- SMOTE **hurts**: AUC dropped 0.10.
- Threshold tuning at 0.164 lifts recall to **40%**.

Choosing the threshold matters more than rebalancing the data

Conclusions

What we learned

- **HistGradientBoosting** wins at val ROC AUC $= 0.7595$ — tree boosting takes the lead by $\sim$1 AUC point.
- **Neural networks** tie LR and RF — depth alone does not unlock new performance on this tabular data.
- **Threshold tuning** on the MLP lifts recall from 0% to 40%; SMOTE hurt AUC.

Limitations

- Only main `application_*.csv` tables — auxiliary tables typically add several AUC points.
- `PredefinedSplit` is fast but optimistic; full k-fold CV would tighten confidence intervals.

Future work

Thank You

Questions?

Recording «INSERT RECORDING LINK BEFORE SUBMISSION»

Code github.com/ymw0414/loan-default-imbalance-classification

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